

# Microprocessors and Microcontrollers Lab

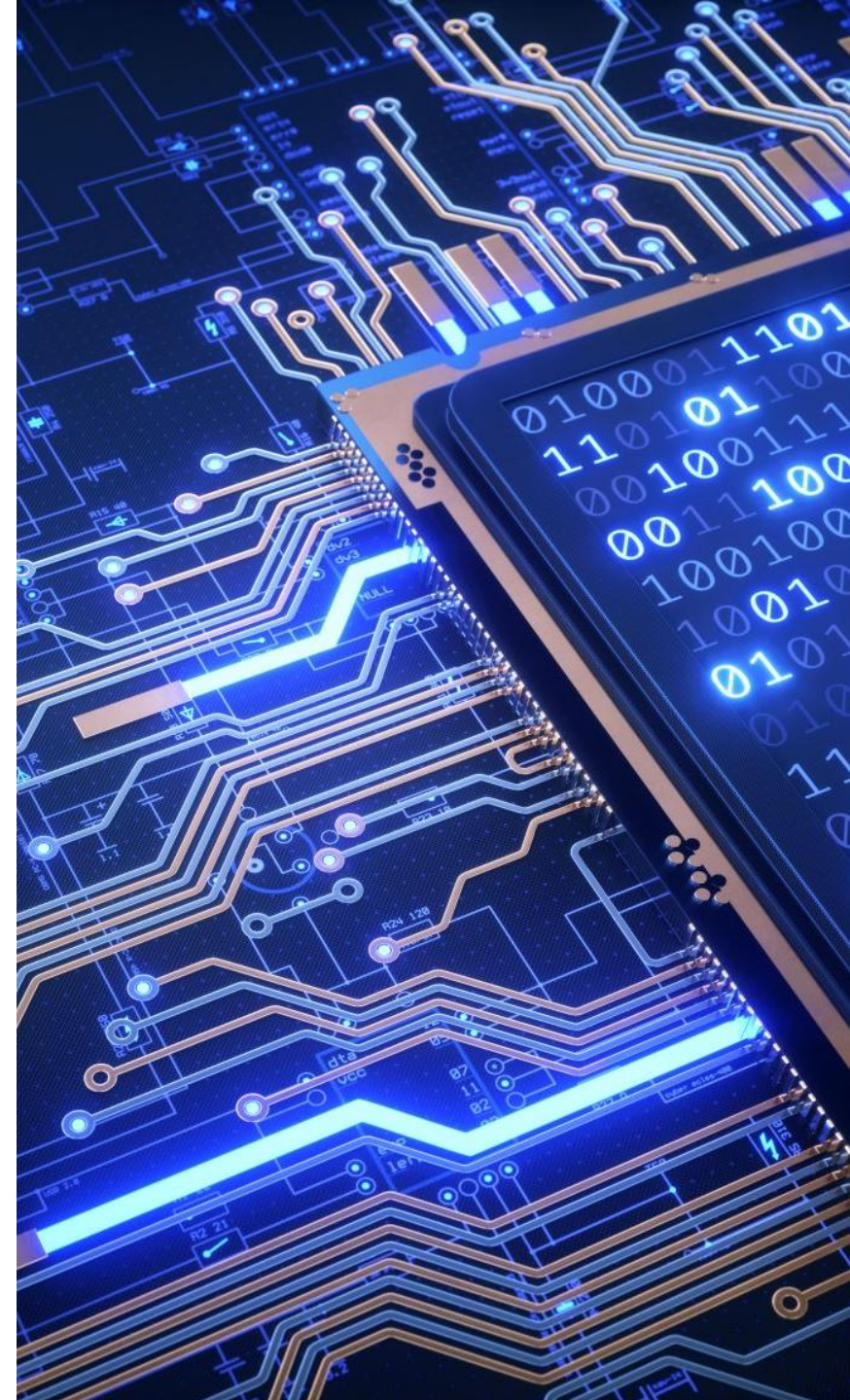
## BECE204P

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# Stack Operation Programming

# Assembly Language Programming

Solution:

Question: Write assembly language programming (ALP) to use the registers of bank 3 and load the value 05H in the registers R0 to R3.

```
ORG 0000H
```

```
SETB PSW.3
```

```
SETB PSW.4
```

```
MOV R0, #05H
```

```
MOV R1, #05H
```

```
MOV R2, #05H
```

```
MOV R3, #05H
```

```
Halt : SJMP Halt
```

```
END
```

# Assembly Language Programming

Question: Write an ALP for loading the data (25H, 12H, F3H) into stack memory from Registers R6, R1, and R4. Assume the default stack area and register bank 0 is selected.

Solution:

```
ORG 0000H  
  
MOV R6, #25H  
MOV R1, #12H  
MOV R4, #0F3H  
PUSH 6  
PUSH 1  
PUSH 4  
Halt : SJMP Halt  
END
```

# Assembly Language Programming

Question: Write an ALP for loading the data (25H, 12H, F3H) into stack memory from Registers R6, R1, and R4. Assume the default stack area and register bank 0 is selected. Also pop out this data from stack memory to the R3, R5, and R2.

Solution:

```
ORG 0000H
MOV R6, #25H
MOV R1, #12H
MOV R4, #0F3H
PUSH 6
PUSH 1
PUSH 4
POP 3
POP 5
POP 2
Halt : SJMP Halt
END
```

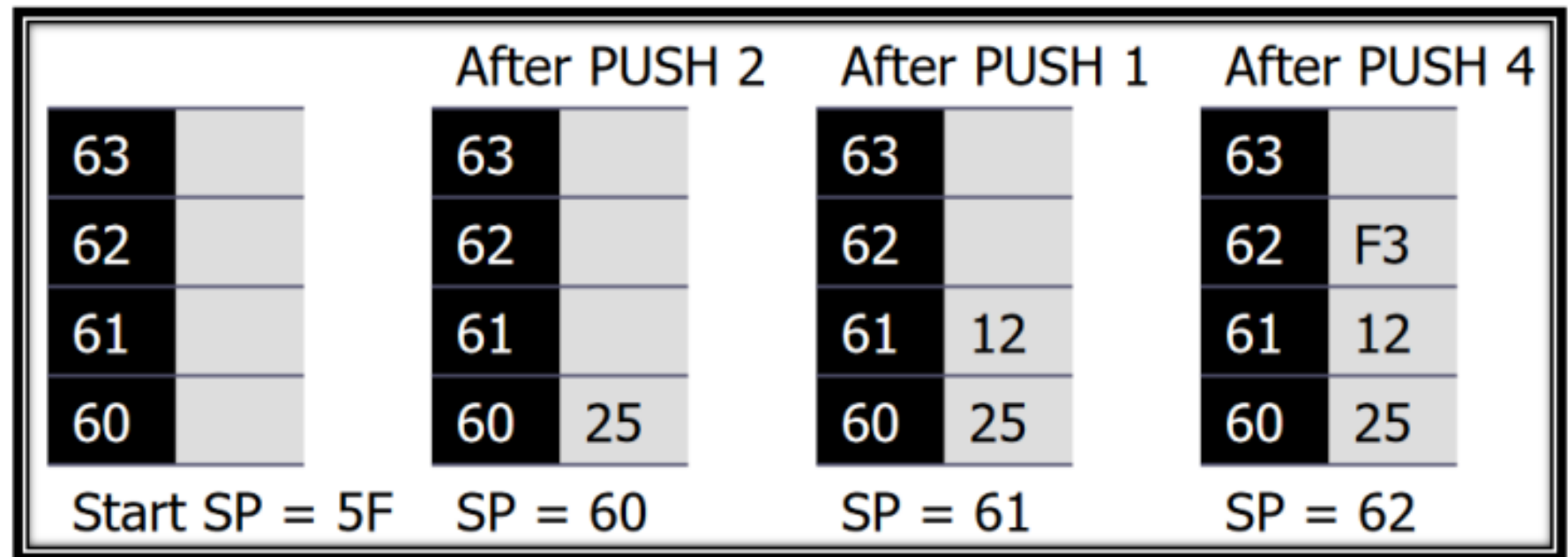
# Assembly Language Programming

Question : Examining the stack, show the contents of the register and SP after execution of the following instructions. All value are in hex.

MOV SP, #5FH ;make RAM location 60H  
;first stack location

MOV R2, #25H  
MOV R1, #12H  
MOV R4, #0F3H

PUSH 2  
PUSH 1  
PUSH 4  
Halt: SJMP Halt



END



# Assembly Language Programming

Question: Write an ALP to load values into each of registers R0 - R4 and then push each of these registers onto the stack and pop them back. Examine the stack and the SP register after the execution of each instruction.      Data: 25H, 35H, 45H, 55H, 65H

Solution:

# Assembly Language Programming

Solution:

ORG 0000H

MOV R0, #25H

MOV R1, #35H

MOV R2, #45H

MOV R3, #55H

MOV R4, #65H

PUSH 0

PUSH 1

PUSH 2

PUSH 3

PUSH 4

POP 4

POP 3

POP 2

POP 1

POP 0

Halt: SJMP Halt

END



# Any Questions?



# Thank you!

